



Nottingham Trent
University

Department of Computer Science

COMP40771 – Computational Intelligence

Lecture 4: BCI and AI for EEG Classification

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Lecture Outline

In this lecture will cover:

- Feature extraction
- Feature selection
- Ensemble learning
- AI for EEG analysis

This lecture contributes to the following learning outcomes:

MLO1 - Demonstrate a comprehensive understanding of the core principles, algorithms, and applications of computational intelligence.

MLO3 - Apply computational intelligence methods to real-world problems and interpret the results effectively

MLO7 - Demonstrate critical thinking and creativity in designing and implementing innovative computational intelligence solutions.

Context and Relevance

- Brain–Computer Interfaces as an application of signal processing and artificial intelligence
- Use of EEG-based BCIs to apply machine learning to real biomedical signals
- Relevance to healthcare and clinical monitoring applications
- Importance in neuroengineering and assistive technologies
- Connection between theoretical methods and real-world bioengineering systems



What is a Brain-Computer Interface?

- System that acquires and interprets brain signals to generate control commands
- Enables direct communication between the brain and an external device
- Operates without relying on peripheral muscles or nerves
- Converts neural activity into machine-interpretable signals
- Can function in open-loop configurations without feedback
- Can operate in closed-loop configurations with real-time feedback and adaptation



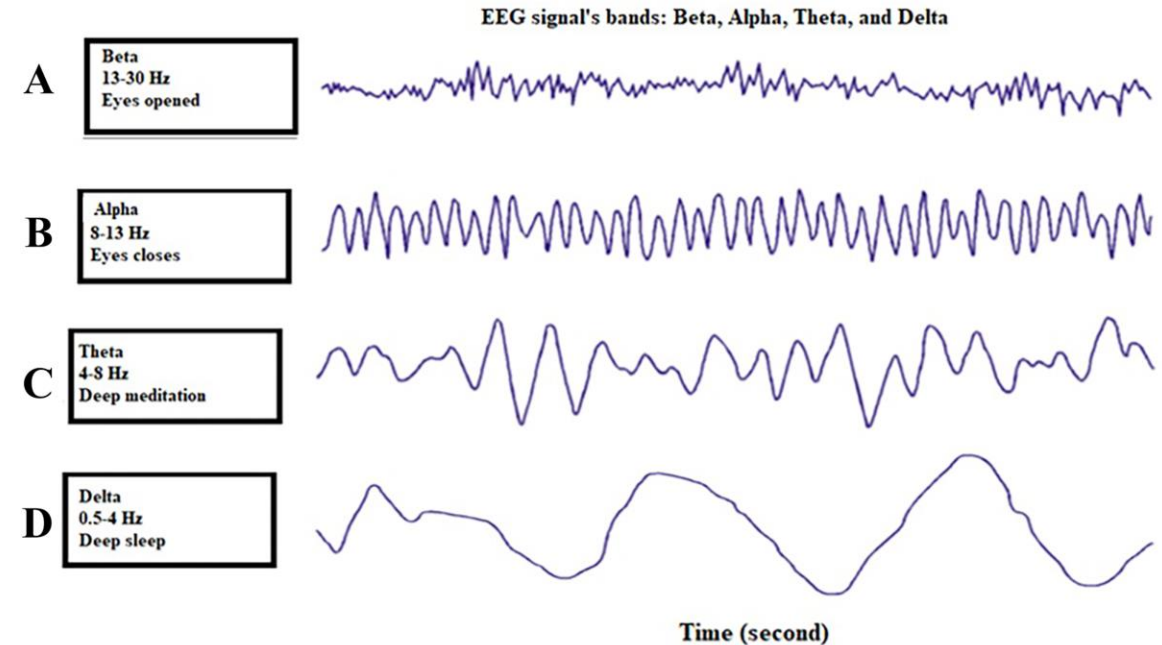
Why EEG for BCI?

- Non-invasive technique that records brain activity from the scalp
- Does not require surgical procedures or implanted electrodes
- Provides high temporal resolution, capturing rapid neural dynamics
- Suitable for real-time BCI applications
- Relatively low cost compared to other neuroimaging modalities
- Widely available in clinical and research environments



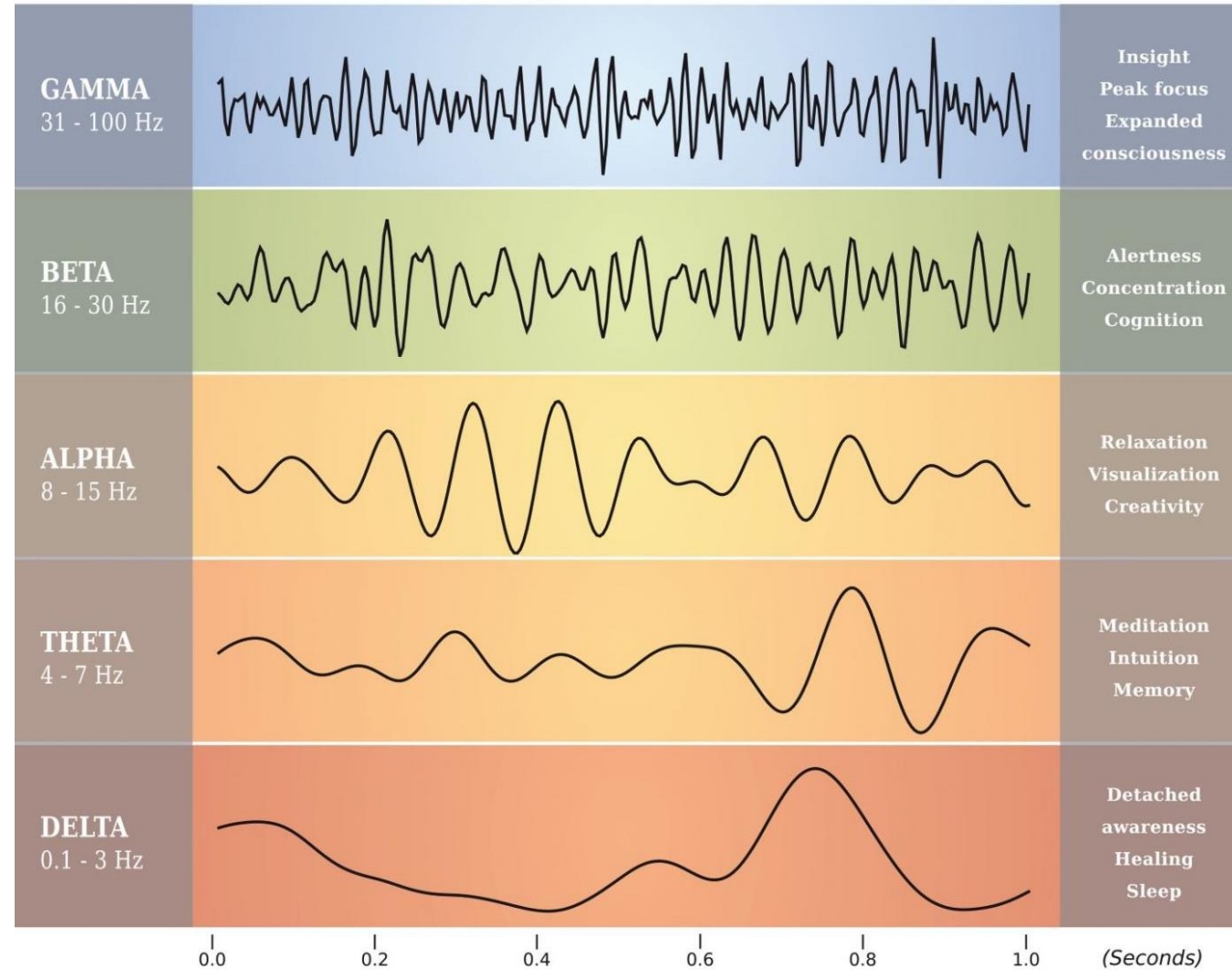
Physiological Origin of EEG Signals

- EEG signals arise primarily from postsynaptic potentials rather than action potentials
- Summation of excitatory and inhibitory postsynaptic activity in pyramidal neurons
- Signals reflect synchronous activity of large neuronal populations
- Cortical pyramidal cells contribute most due to their alignment and geometry
- Recorded signals represent aggregated electrical activity rather than single neurons
- Strong influence of volume conduction through brain tissue, skull, and scalp
- Resulting measurements have low spatial specificity despite high temporal resolution



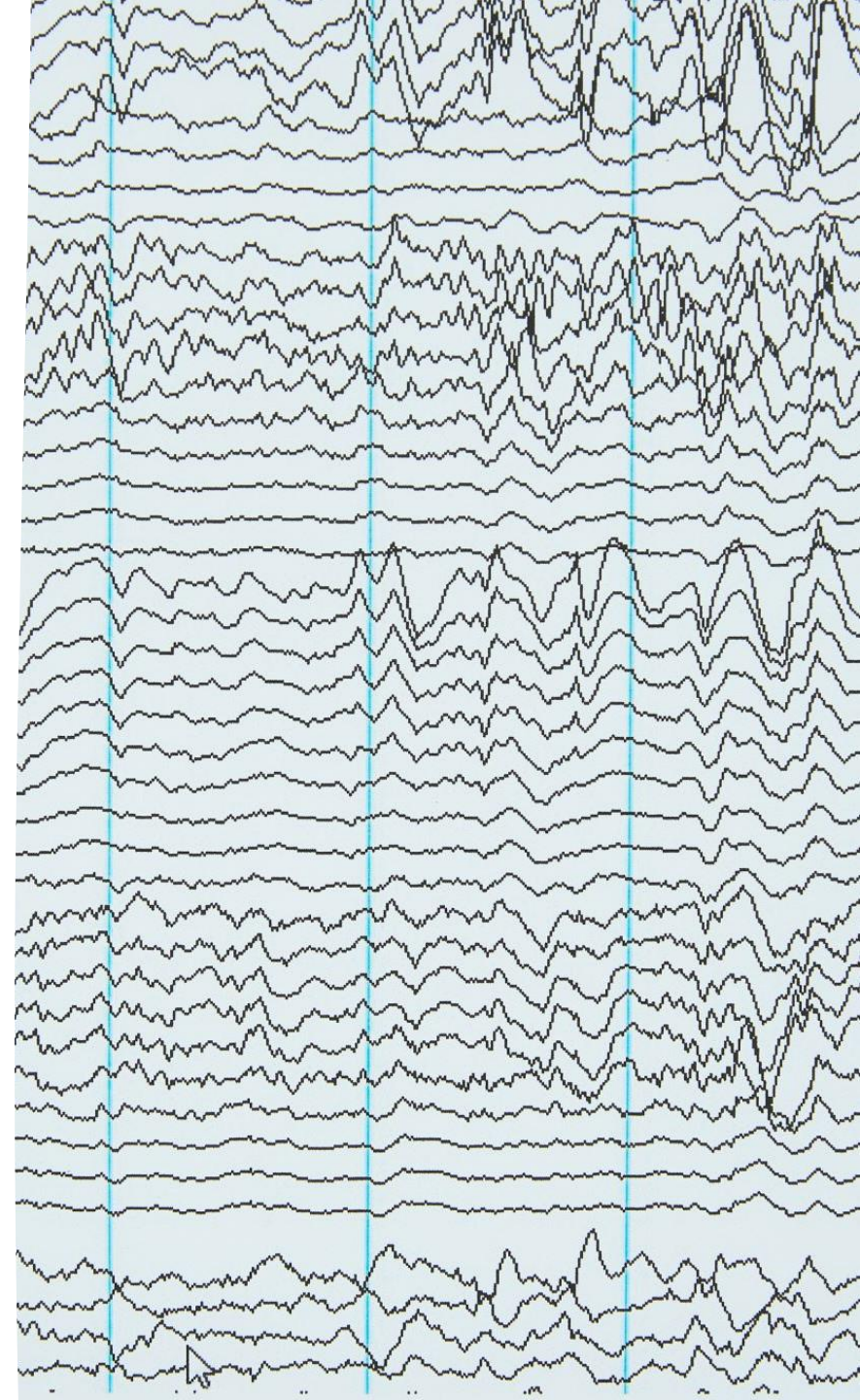
EEG Frequency Bands

HUMAN BRAIN WAVES



Characteristics of EEG Data

- Low signal-to-noise ratio due to weak neural signals and strong external interference
- EEG signals are highly non-stationary, with statistical properties changing over time
- Signal characteristics vary across recording sessions and experimental conditions
- Large inter-subject variability caused by anatomical and physiological differences
- Significant intra-subject variability due to fatigue, attention, and cognitive state
- These properties complicate feature extraction, model training, and generalisation



Noise and Artefacts in EEG

Eye movements and eye blinks producing electrooculographic artefacts

Contamination from facial and neck muscle activity producing electromyographic artefacts

Motion-related artefacts caused by electrode movement and cable displacement

Power-line interference at mains frequency

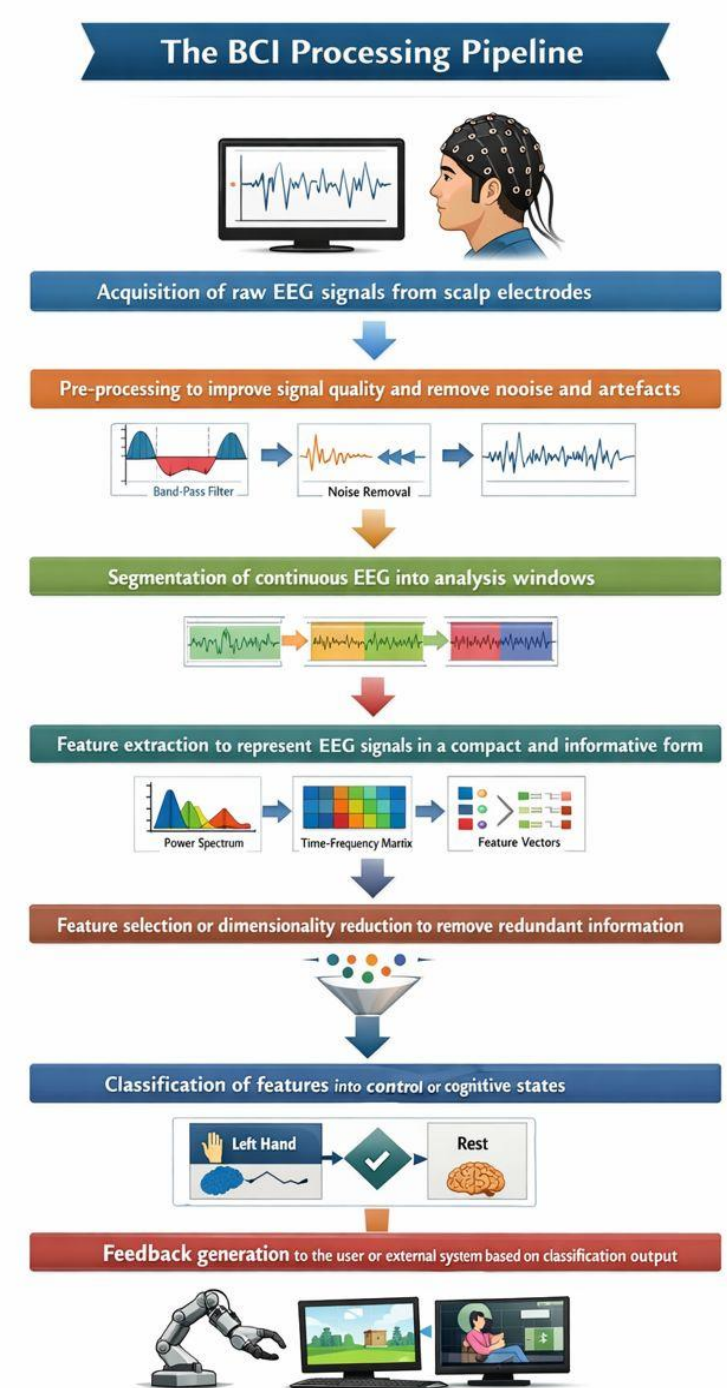
Artefacts often have higher amplitude than neural signals

Effective artefact handling is critical for reliable EEG analysis

Key Challenges in EEG-based BCIs

- Limited availability of labelled EEG datasets for supervised learning
- High cost and time required for data collection and annotation
- Poor generalisation of trained models across different subjects
- Strong subject-specific signal characteristics
- Requirement for real-time or near real-time processing
- Trade-offs between model complexity, latency, and accuracy

The BCI Processing Pipeline



Offline vs Online BCI Systems

- Offline BCI systems used for data analysis, model training, and algorithm development
- Performance evaluated using historical EEG recordings
- Online BCI systems operate in real time during signal acquisition
- Provide immediate feedback or control to the user
- Require low-latency signal processing and decision making
- Use different validation criteria compared to offline systems, including usability and stability

Synchronous vs Asynchronous BCIs

Synchronous BCIs rely on externally defined cues or time windows

- User interaction occurs at predefined moments
- Easier signal segmentation and classifier design
- Generally achieve higher classification accuracy

Asynchronous BCIs allow users to initiate control at any time

- Support self-paced and more natural interaction
- Require continuous monitoring of EEG signals
- Increased difficulty in detecting true user intent
- Trade-off between improved usability and reduced robustness or accuracy

EEG Pre-processing Overview

- Conditioning of raw EEG signals to improve signal quality
- Removal or attenuation of noise and physiological artefacts
- Preparation of data for reliable feature extraction
- Reduction of unwanted variability in the signals
- Strong influence on the performance of downstream machine learning models

Temporal and Spatial Filtering

Application of band-pass filters to retain frequency bands of interest

- Use of notch filters to remove power-line interference
- Suppression of irrelevant or noisy frequency components

Temporal filtering to stabilise EEG signals over time

- Spatial filtering techniques to combine information across electrodes
- Enhancement of signal quality and discriminative patterns for classification

Artefact Handling Strategies

- Manual inspection and rejection of EEG segments contaminated by artefacts
- Identification and removal of artefactual components using Independent Component Analysis
- Separation of neural and non-neural signal sources
- Use of automated algorithms for artefact detection and correction
- Machine learning-based methods to identify artefacts in large datasets
- Aim to preserve neural information while minimising signal distortion

Epoching and Segmentation

Epoching

- Extraction of EEG segments time-locked to specific events or stimuli
- Alignment of neural responses relative to cue or stimulus onset
- Commonly used in event-related potential and task-based paradigms
- Enables comparison of brain responses across trials
- Reduces temporal variability linked to task execution

Segmentation

- Division of continuous EEG into fixed-length analysis windows
- Windows may be non-overlapping or overlapping
- Overlapping segments increase temporal resolution
- Window length controls the trade-off between time resolution and feature stability
- Forms the basis for feature extraction and subsequent classification

Normalisation and Baseline Correction

Normalisation of EEG amplitudes to a common scale

- Reduction of variability caused by electrode impedance and session differences

Baseline correction using pre-stimulus or reference periods

- Improves stability and robustness of extracted features
- Facilitates comparison of EEG data across sessions and subjects

Why Feature Extraction Matters

Raw EEG signals are high dimensional and contain substantial noise

Direct learning from raw signals is inefficient for many classifiers

Feature extraction summarises EEG data into informative representations

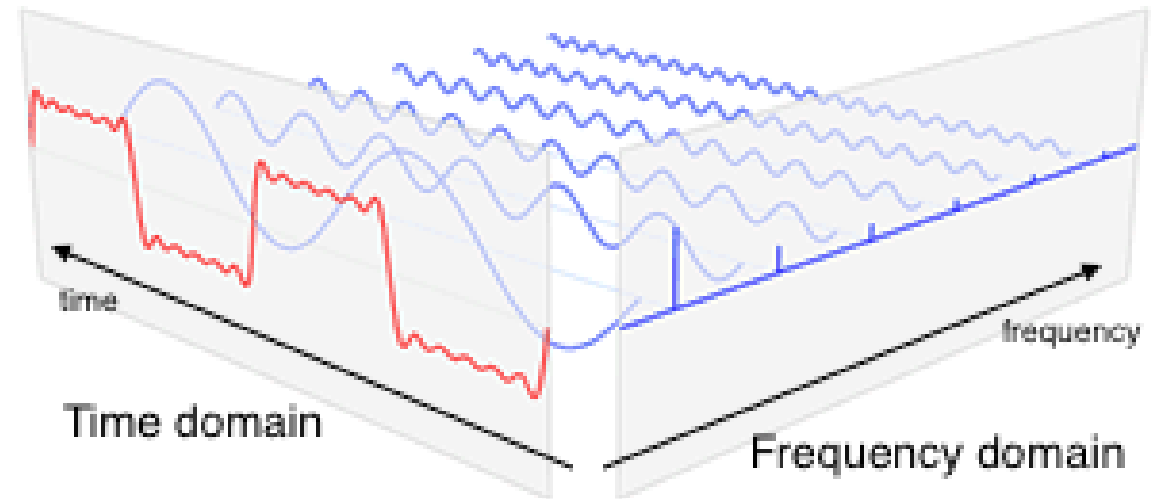
Extracted features highlight discriminative patterns related to brain states

Reduces data dimensionality and redundancy

Improves classifier training, generalisation, and computational efficiency

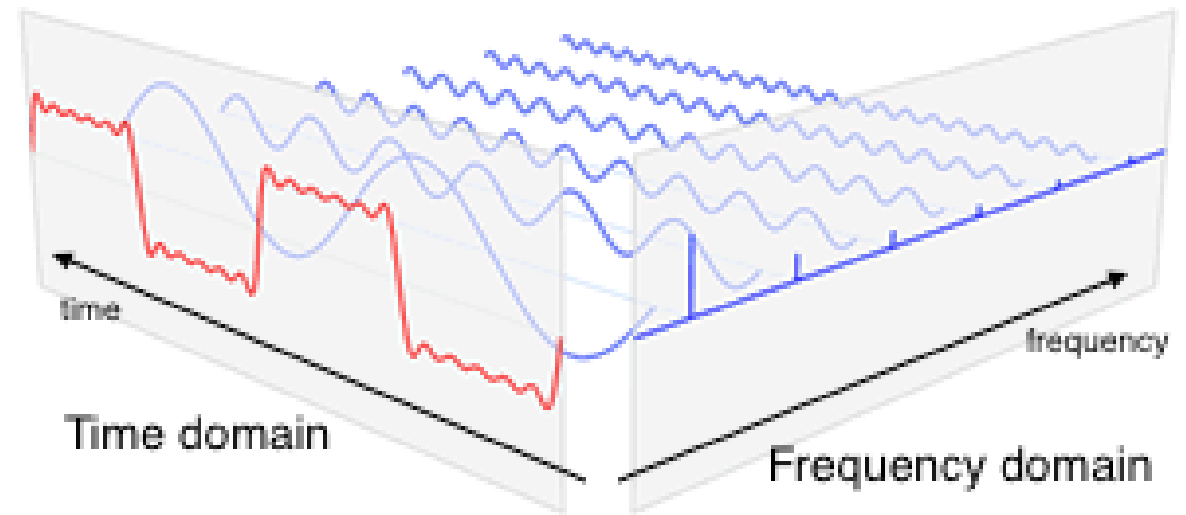
Time-domain EEG Features

- Statistical measures such as mean, variance, and higher-order moments
- Description of signal amplitude and variability over time
- Hjorth parameters capturing activity, mobility, and complexity
- Characterisation of signal shape and temporal dynamics
- Autoregressive coefficients modelling EEG as a linear dynamical process
- Useful for low-complexity and real-time EEG classification tasks



Frequency-domain EEG Features

- Representation of EEG signals in the frequency domain
- Estimation of power spectral density to describe energy distribution across frequencies
- Computation of band power features for standard EEG frequency bands
- Identification of task-related or pathological spectral changes
- Fourier-based analysis methods, such as the Fast Fourier Transform
- Widely used features due to interpretability and robustness



Time-Frequency Representations

- Joint representation of EEG signals in time and frequency domains
- Wavelet transform for multi-resolution analysis of EEG activity
- Effective at capturing transient and non-stationary signal components
- Short-Time Fourier Transform using sliding windows
- Trade-off between time and frequency resolution
- Particularly suitable for analysing event-related and rapidly changing EEG patterns

Spatial Features and CSP Intuition

- Spatial features exploit correlations between multiple EEG channels
- Common Spatial Patterns project EEG signals into a space that maximises variance differences between classes
- Enhances discriminability for binary classification tasks
- Widely applied in motor imagery BCI systems
- Requires careful preprocessing and regularisation
- Sensitive to noise, artefacts, and limited training data
- Prone to overfitting without appropriate feature selection or validation

Non-linear and Complexity-based Features

Features designed to capture non-linear dynamics present in EEG signals

Entropy measures used to quantify signal irregularity and unpredictability

Approximate entropy and sample entropy commonly applied in EEG analysis

Fractal dimension used to characterise signal complexity and self-similarity

Suitable for distinguishing normal and pathological brain states

Effective for analysing non-stationary and chaotic EEG behaviour

Often complementary to time- and frequency-domain features

Hand-crafted vs Learned Features

Hand-crafted features designed using prior domain knowledge of EEG signals

- Examples include statistical, spectral, and spatial features
- Typically interpretable and computationally efficient

Learned features obtained directly from data using machine learning models

- Often capture complex and non-linear patterns automatically
- Reduced need for manual feature design
- Trade-off between higher interpretability of hand-crafted features and potentially higher performance of learned features

Feature Dimensionality and Redundancy

- Large number of extracted features increases the dimensionality of the feature space
- Curse of dimensionality leads to sparse data representation
- Highly correlated or redundant features provide little additional information
- Increased risk of overfitting, especially with small datasets
- Higher computational cost for training and inference
- Necessitates feature selection or dimensionality reduction methods

Motivation for Feature Selection

- Reduction of the size of the feature space
- Removal of irrelevant or redundant features
- Improvement of model generalisation to unseen data
- Reduction of overfitting risk
- Lower computational and memory requirements
- Increased interpretability of learned models

Filter-based Feature Selection

- Selection of features based on statistical properties of the data
- Use of correlation-based measures to identify relevant features
- Application of mutual information to quantify feature–class dependency
- Evaluation performed independently of any classifier
- Computationally efficient and scalable
- Commonly used as an initial feature reduction step

Wrapper-based Feature Selection

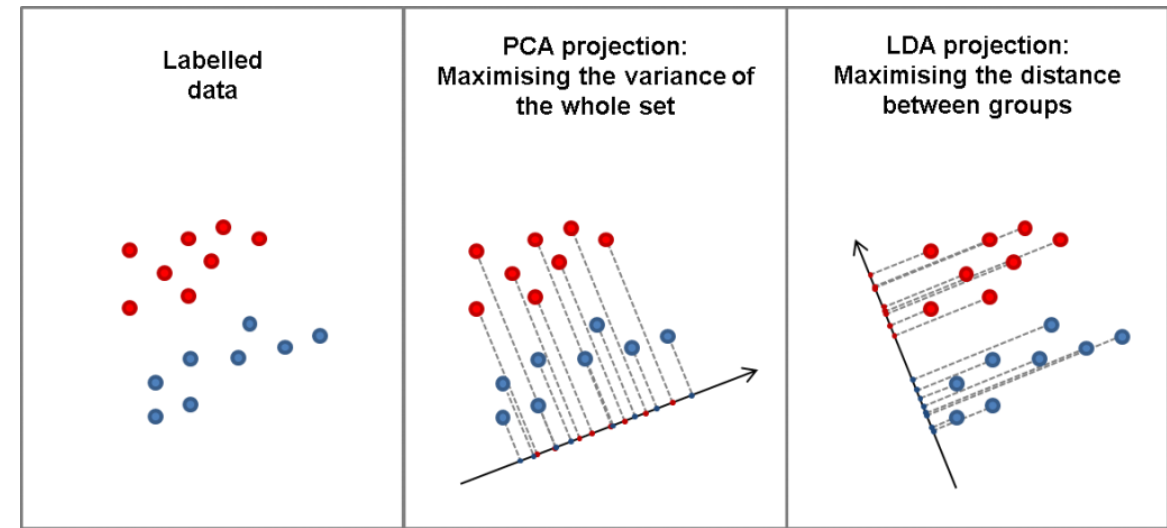
- Feature selection guided by classifier performance
- Evaluation of feature subsets using a chosen learning algorithm
- Recursive feature elimination as a common approach
- Iterative removal of least informative features
- Typically yields high classification performance
- Computationally expensive, especially for high-dimensional data

Embedded Feature Selection

- Feature selection performed as part of the model training process
- Selection driven directly by the learning algorithm
- L1 regularisation promotes sparsity by forcing less relevant feature weights to zero
- Automatic reduction of feature dimensionality during optimisation
- Tree-based models provide feature importance measures
- Importance scores used to rank and select informative features
- Balances performance and computational efficiency

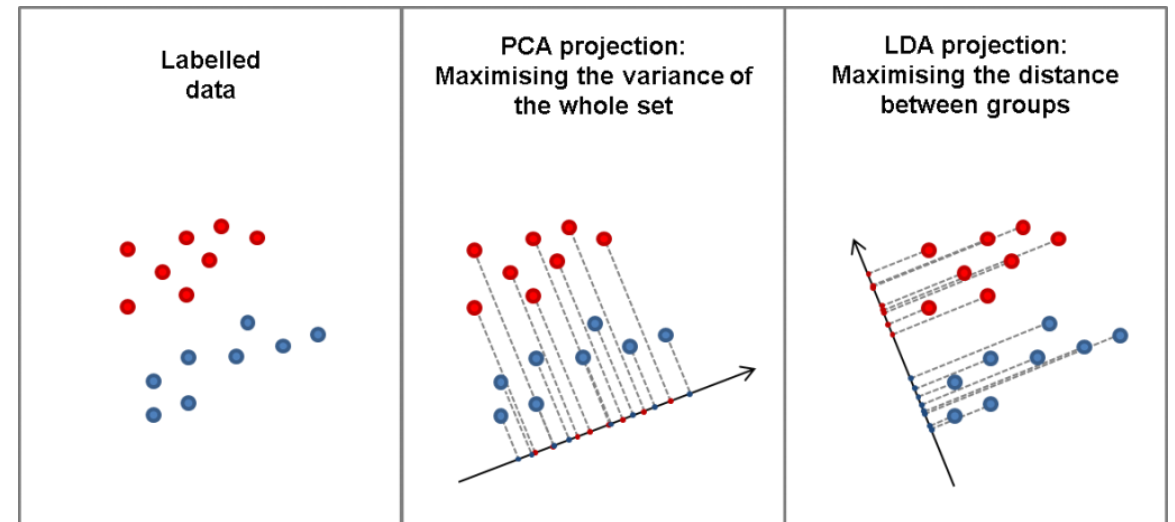
Dimensionality Reduction: PCA

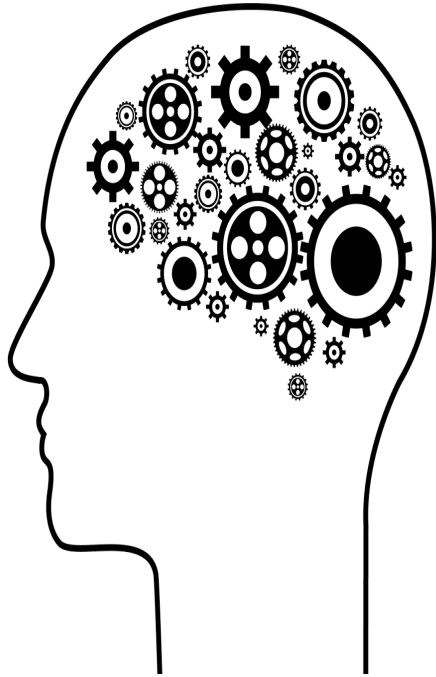
- Unsupervised technique for reducing feature dimensionality
- Projects data onto orthogonal components that maximise variance
- Captures dominant patterns in the data
- Reduces noise and redundancy among correlated features
- Does not use class label information
- May discard discriminative information relevant for classification



Dimensionality Reduction: LDA

- Supervised dimensionality reduction technique
- Uses class labels to guide projection
- Maximises separation between class means
- Minimises variance within each class
- Produces low-dimensional, discriminative feature spaces
- Widely adopted in EEG and BCI classification tasks





Overview of EEG Classification Models

Wide range of classifiers applied to EEG data

Linear classifiers offering simplicity and interpretability

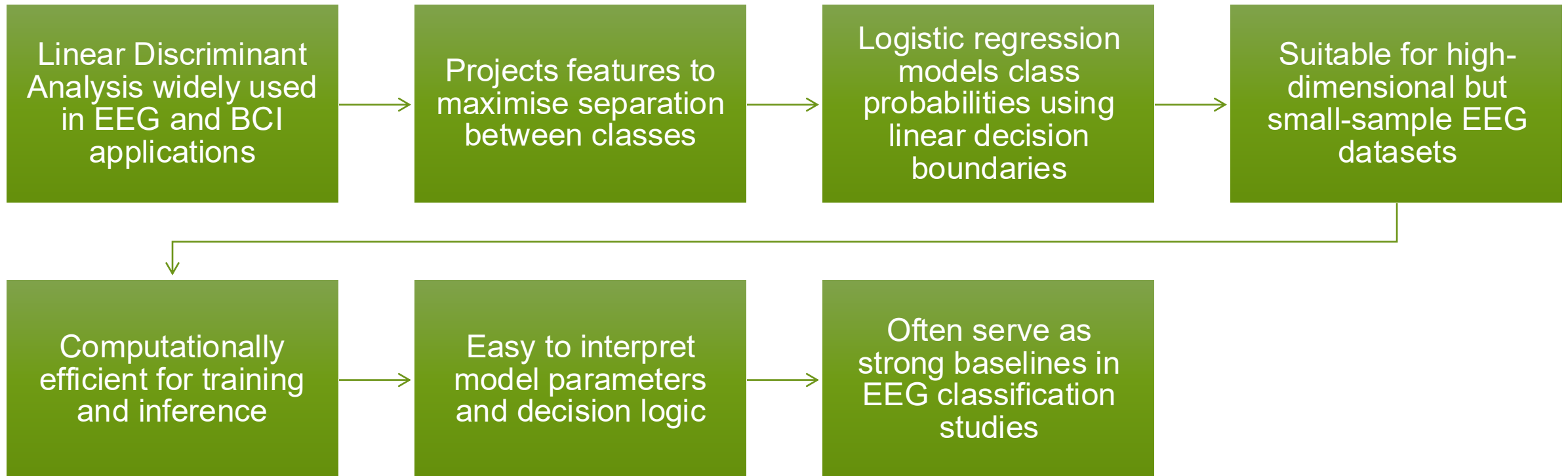
Non-linear classifiers capturing complex decision boundaries

Trade-off between classification accuracy and model complexity

Risk of overfitting with complex models on limited data

Model choice influenced by dataset size, feature type, and real-time constraints

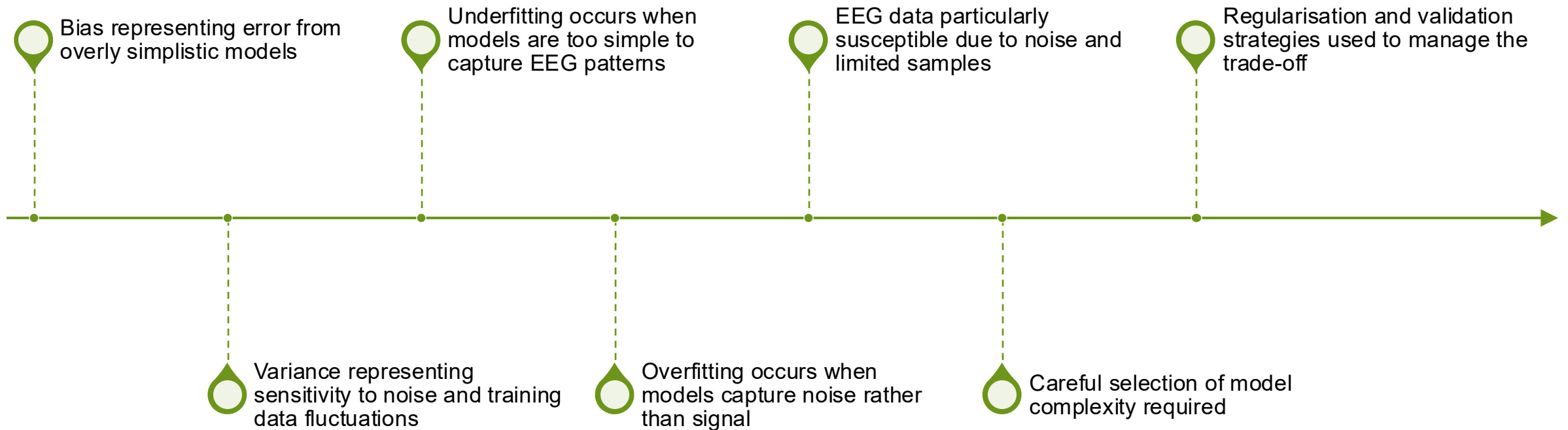
Linear Classifiers for EEG



Non-linear Classifiers for EEG

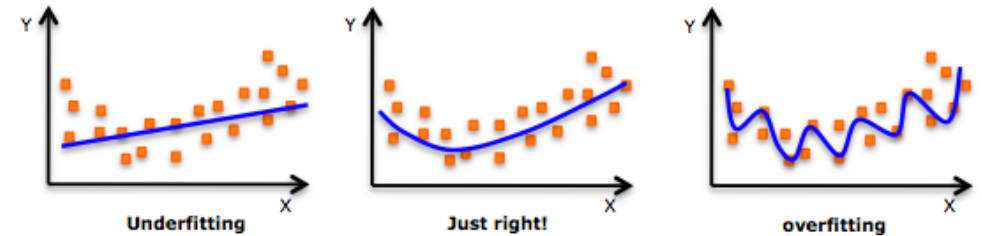
- Support Vector Machines using kernel functions to model non-linear decision boundaries
 - Kernel trick enabling separation of non-linearly separable EEG features
 - Effective performance in many EEG classification tasks with appropriate regularisation
- k-Nearest Neighbours based on distance measures in feature space
 - Simple implementation but sensitive to feature scaling and noise
- Artificial Neural Networks capable of modelling complex, non-linear relationships
 - Flexible architectures suited to diverse EEG feature representations
 - Increased risk of overfitting when applied to small or noisy EEG datasets

Bias/Variance Trade-off

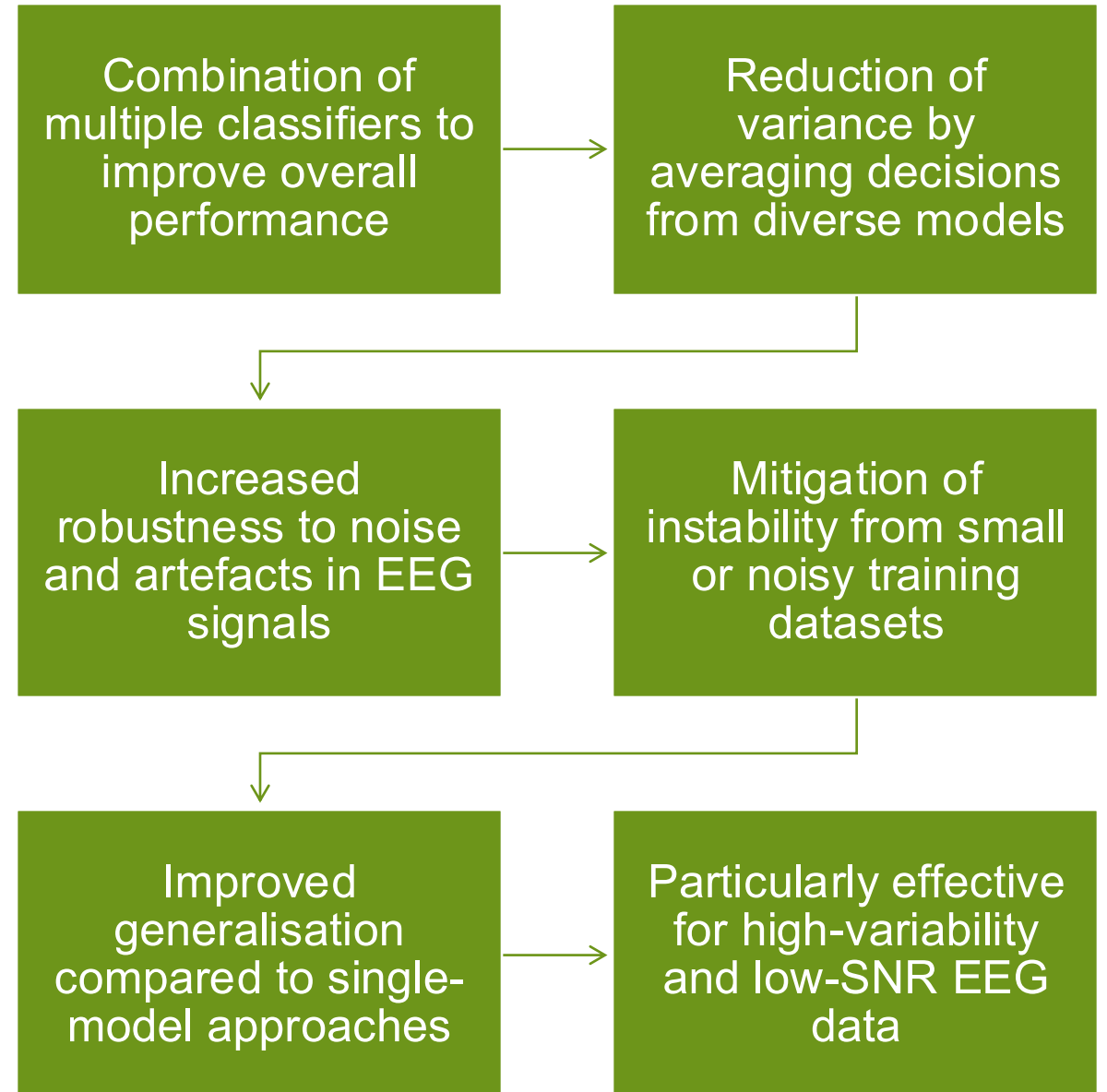


Overfitting and Small-sample Issues

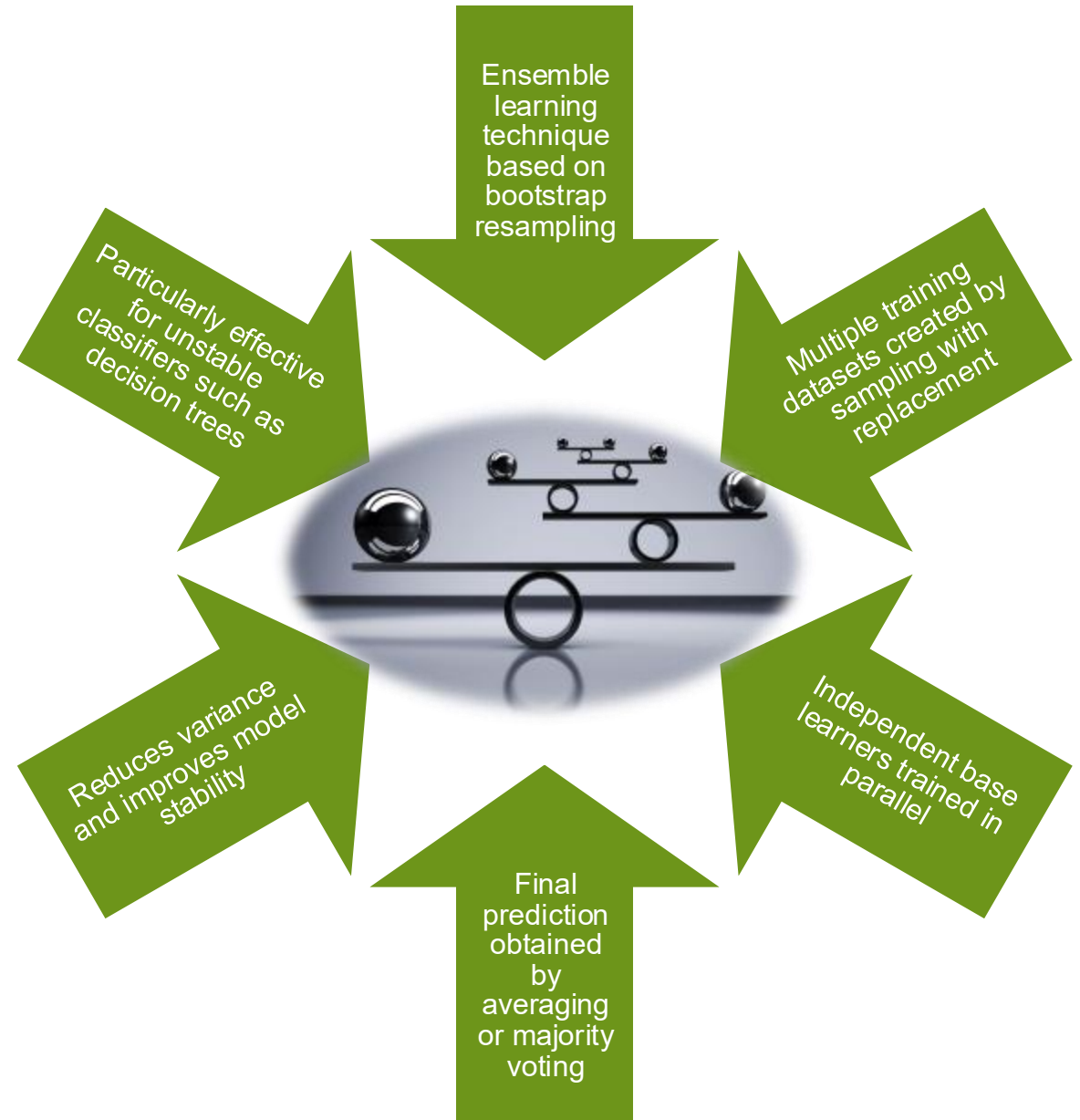
- EEG datasets often contain many features but few training samples
- High feature-to-sample ratio increases overfitting risk
- Models may learn noise rather than meaningful neural patterns
- Limited availability of labelled data for supervised learning
- Regularisation techniques used to constrain model complexity
- Feature selection and dimensionality reduction mitigate overfitting
- Robust validation strategies essential for reliable performance estimation



Motivation for Ensemble Learning



Bagging Methods





Boosting Methods

Ensemble learning approach based on sequential model training

Each learner focuses on samples misclassified by previous models

Training instances weighted to emphasise difficult cases

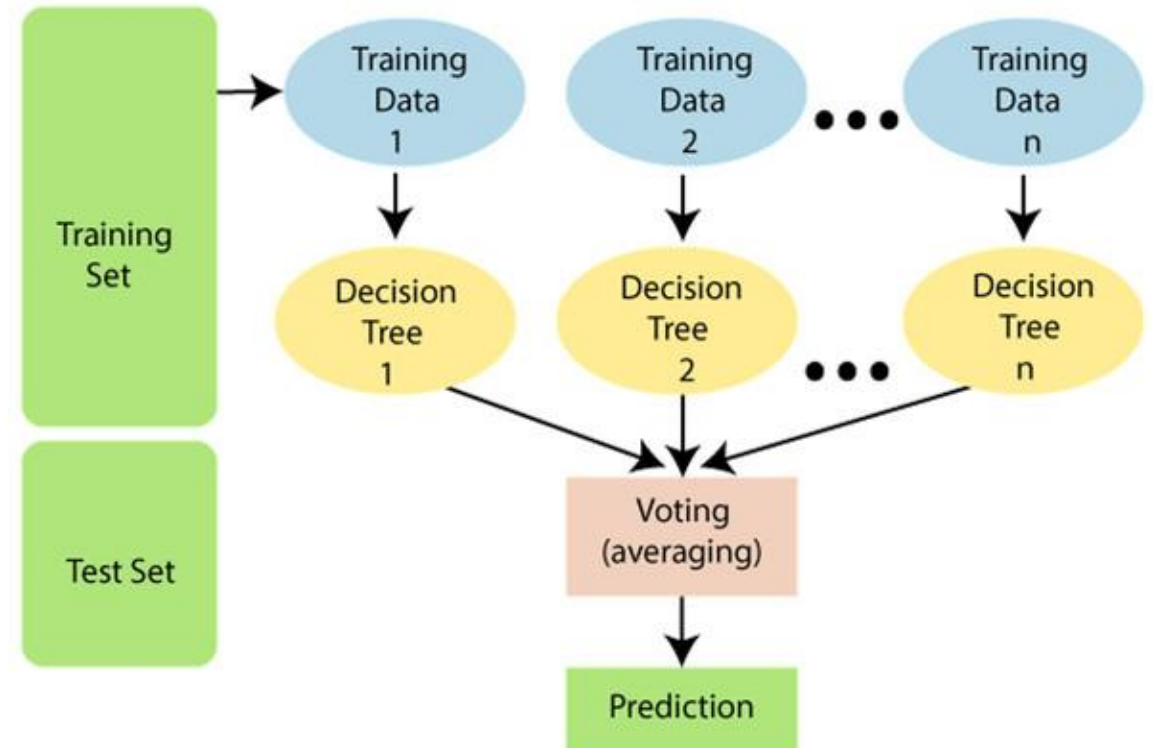
Combination of weak learners into a strong classifier

Often achieves higher accuracy than individual models

Increased sensitivity to noise and outliers compared to bagging

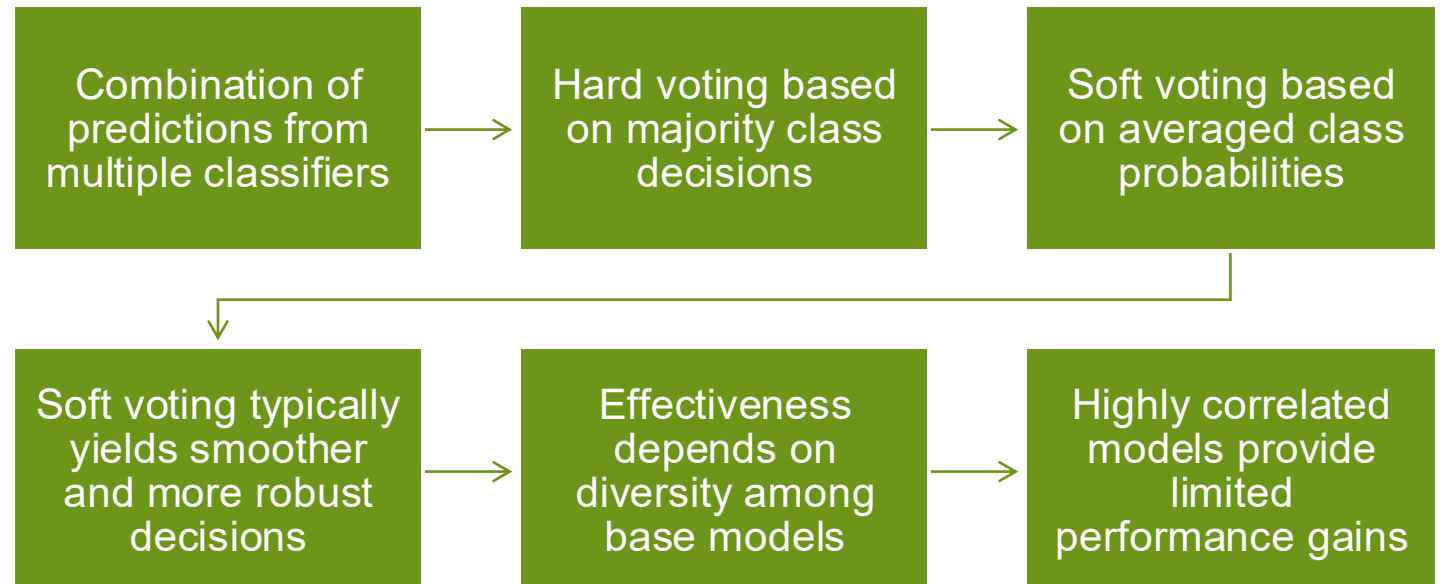
Random Forests for EEG

- Ensemble method composed of multiple decision trees
- Use of bootstrap sampling to create diverse training subsets
- Random selection of feature subsets at each split
- Combination of bagging and feature randomness
- Reduces overfitting compared to single decision trees
- Provides built-in estimates of feature importance
- Demonstrated strong empirical performance in many EEG classification studies





Voting and Averaging Ensembles



Stacking and Hybrid Ensembles

Ensemble approach using a meta-classifier to combine base learners

- Base models may be heterogeneous, using different algorithms or feature sets

Meta-classifier learns how to weight and combine predictions

- Capable of capturing complementary strengths of different models
- Often improves generalisation performance
- Increased complexity and risk of overfitting without careful validation

Model Evaluation in BCI

Offline evaluation based on pre-recorded EEG datasets

- Use of accuracy and related metrics to compare algorithms
- Cross-validation to estimate generalisation performance

Online evaluation focusing on real-time system behaviour

- Assessment of latency, stability, and error rates during use
- Consideration of user adaptation and learning effects
- Inclusion of user-centred metrics such as usability and fatigue

Accuracy and Class Imbalance

Standard accuracy can be biased when class distributions are uneven

- Imbalanced classes are common in EEG and BCI datasets
- Balanced accuracy compensates by averaging performance across classes

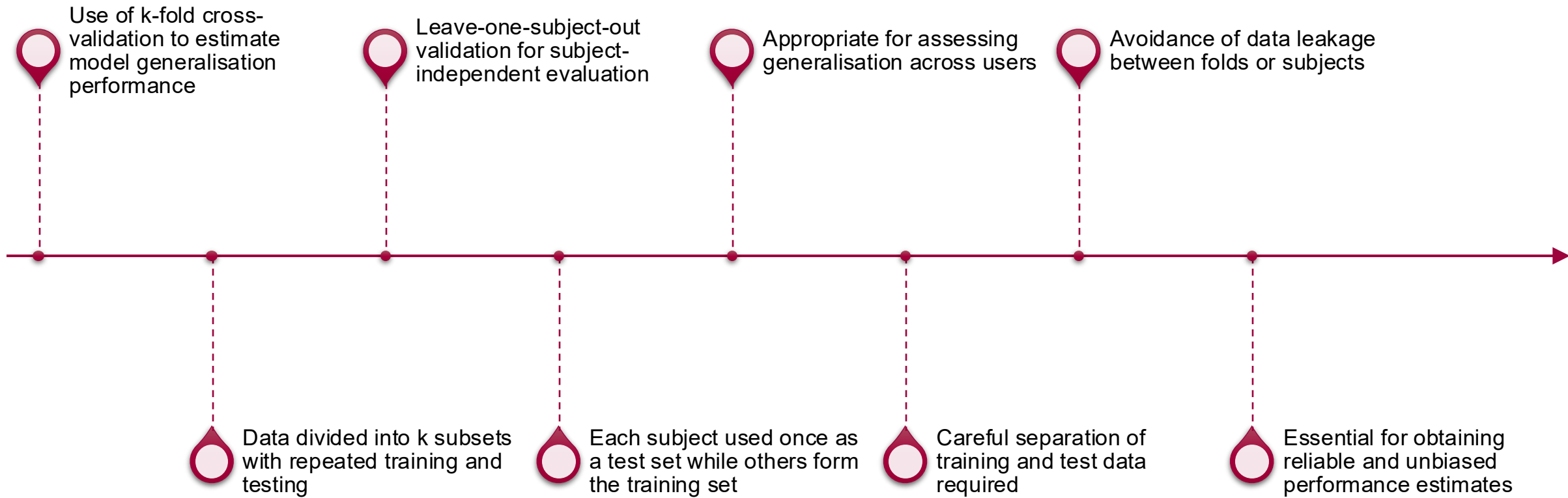
Confusion matrices provide a detailed breakdown of classification outcomes

- Analysis of true positives, false positives, and false negatives

Class-wise performance metrics highlight strengths and weaknesses of models

- Essential for reliable and fair evaluation of BCI systems

Cross-validation Strategies



Case Study: EEG Classification Pipeline

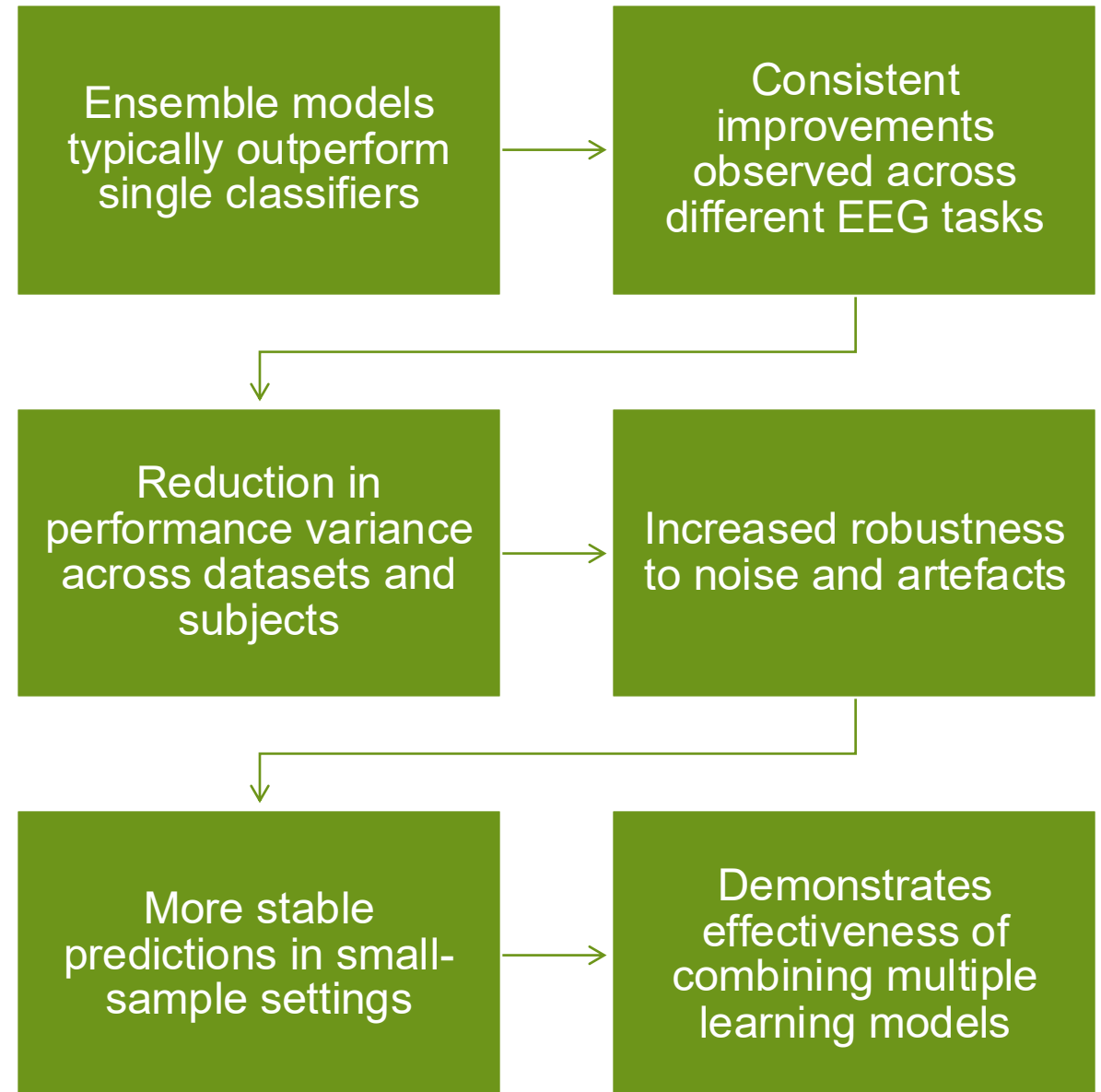
- Description of an end-to-end EEG classification workflow
- Selection of pre-processing steps based on signal characteristics
- Application of appropriate filtering and artefact removal methods
- Extraction of time, frequency, or spatial features
- Feature selection or dimensionality reduction to control complexity
- Training of multiple classification models
- Comparison of model performance using consistent evaluation metrics

Feature Extraction and Selection Choices

- Choice of features strongly influences classification performance
- Different feature types capture complementary aspects of EEG signals
- Feature selection affects model accuracy and robustness
- Trade-off between complex features and interpretability
- Simpler features often easier to interpret clinically
- Computational cost varies with feature dimensionality
- Practical systems balance accuracy, interpretability, and efficiency



Ensemble Learning Results



Interpretation of Performance



Evaluation of model strengths in relation to specific EEG tasks



Identification of limitations such as overfitting or sensitivity to noise



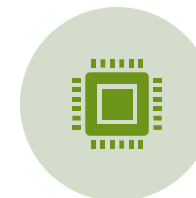
Performance depends strongly on data quality and experimental design



Conclusions must be interpreted within the application context



Trade-offs between accuracy, latency, and complexity



Consideration of robustness and reliability for real-world deployment

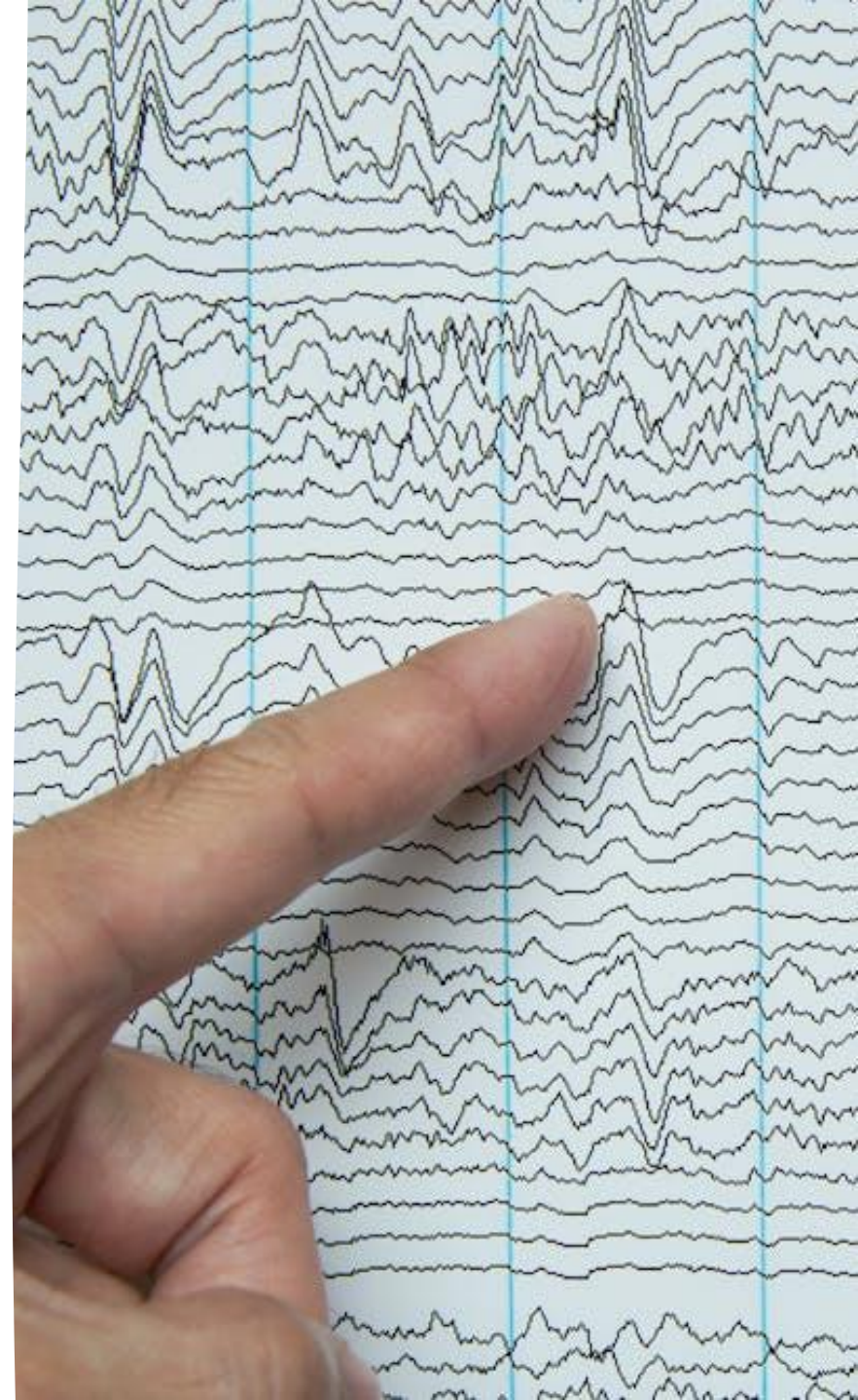
Open Challenges in EEG-based BCIs

- Difficulty in generalising models across different users
- Strong subject-specific EEG characteristics
- Need for adaptation to changes in brain signals over time
- Performance degradation due to fatigue or learning effects
- Scalability to larger populations and longer-term use
- Ensuring robustness in real-world and uncontrolled environments



Current Research Trends

- Increasing use of deep learning models for EEG analysis
- Automatic feature learning from raw or minimally processed signals
- Development of hybrid systems combining classical and deep models
- Continued use of ensemble methods to improve robustness
- Integration of EEG with other modalities such as Functional near-infrared spectroscopy (fNIRS) or Electromyography (EMG)
- Multimodal data fusion for improved classification performance



Key Take-Away

- EEG signals are challenging to analyse due to noise and variability
- Despite challenges, EEG provides rich information about brain activity
- Careful feature extraction and selection are critical for good performance
- Model choice must balance complexity and generalisation
- Ensemble learning methods effectively improve robustness and accuracy
- Reliable BCI systems require thoughtful integration of signal processing and machine learning



Quiz (1/3)

Q1. Why is feature extraction necessary when working with raw EEG signals?

- A. EEG signals are already low-dimensional
- B. Raw EEG is noisy and high dimensional
- C. Classifiers cannot process time-series data
- D. Feature extraction removes the need for preprocessing

Q2. Which property of EEG data most strongly motivates the use of ensemble learning?

- A. High spatial resolution
- B. Deterministic signal behaviour
- C. High inter-subject variability and noise
- D. Large labelled datasets

Quiz (2/3)

Q3. Which feature selection approach evaluates feature subsets using classifier performance?

- A. Filter-based selection
- B. Embedded selection
- C. Wrapper-based selection
- D. Principal Component Analysis

Q4. Which statement best describes the difference between PCA and LDA?

- A. Both are supervised methods
- B. PCA maximises variance, LDA maximises class separability
- C. LDA does not use class labels
- D. PCA is only used for classification

Quiz (3/3)

Q5. Why are linear classifiers such as LDA often strong baselines for EEG classification?

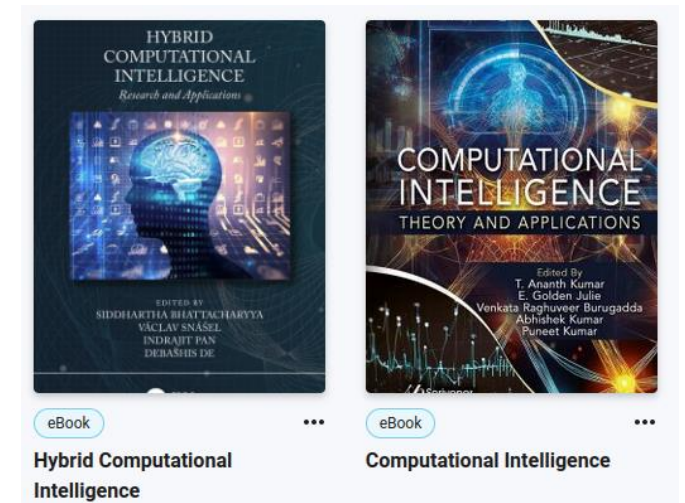
- A. EEG signals are linear by nature
- B. They require no feature extraction
- C. They are efficient and robust for small datasets
- D. They always outperform non-linear models

Q6. Which factor most strongly increases the risk of overfitting in EEG classification?

- A. Low temporal resolution
- B. High feature-to-sample ratio
- C. Use of cross-validation
- D. Ensemble learning

References

[1] Hosseini, M.-P., Hosseini, A., & Ahi, K. (2021). A Review on Machine Learning for EEG Signal Processing in Bioengineering. IEEE Reviews in Biomedical Engineering, 14, 204–218. doi:10.1109/RBME.2020.2969915





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Summary

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- Feature selection
- Ensemble learning
- AI for EEG analysis

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